Aadittya Tiwari

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Objective

Electrical Engineering third-year with first author research experience in Optoelectronic Finite Element Analysis simulations and Quantum Machine Learning model design looking for Spring or Summer 2027 internships.

Education

Georgia Institute of Technology - B.S, Electrical Engineering. 3.60 GPA *Jan. 2026 – May 2028*

University of California, Santa Cruz - Transferred out with a 3.95 GPA *June 2024 – Dec. 2025*

Research Work Experience

Modeling optical loss in Optical Quantum Computers with COMSOL *Jan. 2026 – Oct. 2026*

First Author, Georgia Tech Quantum Computing Club (Club website)

- Tasked with modeling a 3 qubit linear optical quantum computer in COMSOL to optimize design and minimize optical loss, enabling the club's goal of assembling a fully student-built quantum computer
- Ran parameter sweep over laser wavelength, beamwidth, refractive index change by temperature, and organizational distance between optical components to characterize optical loss response of LOQC system
- Extended upon a polarizing beamsplitter public model by implementing phase matching across prism-air interfaces, adding antireflection coating, and 1/3 beamsplitters to recreate an interferometric Control-Z gate
- Wrote and presented grants to Georgia Tech funding sources to purchase optical components, awarded $14,000
- Presented as a poster and paper at COMSOL 2026 Boston Conference in October

Optimizing Electric Field Distributions of Nanopores with COMSOL *Nov. 2024 – Dec. 2025*

Third Author, University of California, Santa Cruz (Applied Optics Group)

- Tasked with recreating exact nanopore chip structure used by PhD student in lab within COMSOL for finite element analysis of the electric field and corresponding particle capture radius
- Eliminated out of memory errors by creating custom mesh with varying element sizes depending on proximity to regions of interest and editing COMSOL's internal files to increase access to RAM from 4 to 13 Gigabytes
- Automated parameter sweep to vary the amount of nanopores (1,2,4,8,10) and distance between nanopores (100 nanometers to 6 micrometers in 100 nm increments) to calculate capture radius for each configuration
- Exported raw electric field distribution data from COMSOL to MATLAB to find an equation to model particle capture radius as a logistic function of nanopore count and placement for future regression tasks
- Paper in review for journal publication with ACS Nano

Quantum Machine Learning for Time Series Forecasting *June 2023 – Sept. 2025*

First Author, University of California, Los Angeles (TensorWorks Lab)

- Tasked with applying my experience with Quantum Machine Learning (QML) models to help a PhD Student with analyzing advantages/disadvantages of using QML in industrial turbine degradation forecasting
- Architected a model that uses a Quantum Long Short Term memory (QLSTM) as the Generator and classical Convolutional Neural Network as discriminator within a Generative Adversarial Network (GAN)
- Implemented Quantum GAN to forecast turbine's vibration signals, converging in 2.7x less epochs (49 classical epochs, 18 quantum) with only 169 QLSTM parameters VS 8,186,113 classical LSTM parameters.
- Deployed repository to a High Performance Computing cluster using SLURM to parameterize train/test split across 5 values and repeat each split 10 times to quantify variance in each model's RMSE performance
- Paper accepted and presented to ESREL'24 conference Summer 2024. Currently revising conference paper for journal submission, code is open-sourced on github with 14 stars and 5 forks

Skills and Relevant Coursework

Languages & Frameworks: Python, Pytorch, Pennylane, Qiskit, Pandas, Numpy, MATLAB, C

Technologies: COMSOL, SLURM, VS Code, Cursor, Git, Supabase, Microsoft Office

Relevant Coursework: Optoelectronics & Photonics, Laser Theory & Applications, Modern Physics